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1 On manifolds with boundary

In this section we discuss how known theorems/lemmas fail for manifolds with non-empty boundary and how they can be generalized.

1.1 Regular values and transversality

It is known that for the regular value theorem, and more generally statements about transversality, to still hold for manifolds with boundary one needs to impose additional conditions on the boundary behaviour of a smooth function $f: M \rightarrow N$. Here is another lemma, that fits into the same category. We call $q \in N$ a **boundary value** of f if $q \in f(\partial M)$.

Lemma 1.1. *Let M be a manifold with boundary and $f: M \rightarrow \mathbb{R}$ a smooth function. Furthermore, let $a \in \mathbb{R}$ be a regular value of f , that is not a boundary value. Then $M_a := f^{-1}(-\infty, a]$ is a submanifold with boundary*

$$\partial M_a = f^{-1}(a) \cup (M_a \cap \partial M).$$

Proof sketch. For $p \in f^{-1}(a)$ we have $p \in \text{int}(M)$ by assumption. Since $\text{int}(M)$ is a boundaryless manifold (and an open subset of M) the local submersion theorem holds, i.e. f looks like the orthogonal projection $\mathbb{R}^n \rightarrow \mathbb{R}$ around p . Then the proof goes essentially the same as in the boundaryless case. $\square$

Counterexample 1.2. To see that Lemma 1.1 fails without the assumption that a is not a boundary value consider $M := S^1 \times [0, 1]$ and $f: M \rightarrow \mathbb{R}, (z, t) \mapsto t$. Then f is everywhere regular but $f^{-1}(-\infty, 0]$ does not satisfy the statements in Lemma 1.1.

Let M, N be manifolds with boundary and $f: M \rightarrow N$ smooth. We define the **boundary map** ∂f of f by

$$\partial f := f|_{\partial M}.$$

We now generalize two theorems from [?] about transversal maps by allowing the codomain to have boundary as well.

Theorem 1.3 (Transversality Theorem). *Let M, N be manifolds with boundary, $f: M \rightarrow N$ smooth and $X \subseteq N$ a submanifold without boundary. If*

$$f \pitchfork X \text{ and } \partial f \pitchfork X$$

then $X' := f^{-1}(X)$ is a submanifold with boundary

$$\partial X' = X' \cap \partial M$$

and

$$\text{codim}(X') = \text{codim}(X).$$

Proof. The case $\partial N = \emptyset$ is shown in [?] (p. 60). If $\partial N \neq \emptyset$ consider the inclusion

$$\iota: N \rightarrow D(N),$$

where $D(N)$ denotes the double of N and define $F := \iota \circ f$. Then $Y := \iota(X)$ is a submanifold of $D(N)$ and $X' = F^{-1}(Y)$. Furthermore, since ι is an embedding (and $\dim(N) = \dim(D(N))$) it follows that

$$F \pitchfork Y \text{ and } \partial F \pitchfork Y,$$

such that the statement follows from the boundaryless case. $\square$

Theorem 1.4 (Parametric Transversality Theorem). *Let M, N be manifolds with boundary and $X \subseteq N$ a submanifold without boundary. Furthermore, let S be a manifold without boundary and $F: M \times S \rightarrow N$ a smooth map with*

$$F \pitchfork X \text{ and } \partial F \pitchfork X.$$

Then for almost every $s \in S$ the map $f_s: M \rightarrow N$ defined by $f_s(p) := F(p, s)$ satisfies

$$f_s \pitchfork X \text{ and } \partial f_s \pitchfork X.$$

Proof. By Theorem 1.3 we know that

$$X' := F^{-1}(X)$$

is a submanifold (with boundary) of $M \times S$. Consider the projection $\pi: M \times S \rightarrow S$. We can then show, that for a regular value $s \in S$ of the map $\pi|_{X'}$, the maps f_s and ∂f_s are transversal to X , which is just some linear algebra. The proof is exactly the same as in the case $\partial N = \emptyset$, see [?] (p. 68). Then the statement follows from Sard's Theorem. $\square$

Theorem 1.5 (Transversality Homotopy Theorem). *Let M, N be manifolds with boundary, $f: M \rightarrow N$ smooth and $X \subseteq N$ a submanifold without boundary. Then f is smoothly homotopic to a map $g: M \rightarrow N$ with*

$$g \pitchfork X \text{ and } \partial g \pitchfork X.$$

Proof. First we may assume that $\text{im}(f) \subseteq \text{int}(N)$, since f is always smoothly homotopic to such a map (the inclusion $\text{int}(N) \hookrightarrow N$ is a smooth homotopy equivalence). Define

$$Y := X \cap \text{int}(N)$$

and

$$f': M \rightarrow \text{int}(N), \quad p \mapsto f(p).$$

Using the Transversality Homotopy Theorem for the case $\partial N = \emptyset$ (see [?] p. 70) we get that f' is smoothly homotopic to a map $g': M \rightarrow \text{int}(N)$, such that

$$g' \pitchfork Y \text{ and } \partial g' \pitchfork Y.$$

Now define $g := \iota \circ f'$, where $\iota: \text{int}(N) \hookrightarrow N$ is the inclusion. Since g does not intersect X at boundary points, i.e. $\text{im}(g) \cap X \subseteq \text{int}(N)$, the transversality properties of g' directly imply

$$g \pitchfork X \text{ and } \partial g \pitchfork X.$$

$\square$

1.2 Morse Theory

For a smooth function $f: M \rightarrow \mathbb{R}$ and a regular value $a \in \mathbb{R}$ that is not a boundary value we define the **sublevel set** of a to be

$$M_a := f^{-1}(-\infty, a].$$

Lemma 1.1 guarantees, that M_a is a (sub-)manifold with boundary.

Lemma 1.6. *Let M be a compact manifold with boundary and $f: M \rightarrow \mathbb{R}$ a Morse function. Furthermore let $a, b \in \mathbb{R}$ be regular values of f , such that $[a, b]$ does not contain any critical values nor any boundary values. Then*

$$M_a \cong M_b.$$

Proof. Let $K := f^{-1}[a, b]$ and let $U := \{p \in \text{int}(M) \mid p \text{ is a regular point of } f\}$. Then $U \subseteq M$ is open and $K \subseteq U$. Choose a cutoff function $\phi: M \rightarrow \mathbb{R}$ with

- $\phi|_K = 1$
- $\text{supp}(\phi) \subseteq U$
- $0 \leq \phi \leq 1$

Now choose a pseudo gradient¹ X of f and define the smooth vector field

$$Y := \phi \frac{-1}{X \cdot f} X,$$

¹Here we agree to the convention that $X \cdot f < 0$ outside of critical points.

which we define to be 0 outside of U . Since Y vanishes in a neighbourhood around ∂M (or more generally since Y is tangent to ∂M , see [?] Theorem 9.34) we can consider the flow of Y :

$$\theta: M \times \mathbb{R} \rightarrow M.$$

Since M is compact (and Y tangent to ∂M) each integral curve is defined for all time. Let $\theta_t := \theta(-, t): M \rightarrow M$. We claim that

$$\theta_{b-a}(M_b) \subseteq M_a, \quad (1)$$

$$\theta_{a-b}(M_a) \subseteq M_b. \quad (2)$$

Let $p \in f^{-1}[a, b]$ and let $\gamma_p: \mathbb{R} \rightarrow M$ be the integral curve of Y starting at p . Consider

$$h: \mathbb{R} \rightarrow \mathbb{R}, \quad t \mapsto f(\gamma_p(t)).$$

Then for $t \in \mathbb{R}$ with $h(t) \in [a, b]$ we have $\phi(\gamma_p(t)) = 1$ and thus

$$h'(t) = \dot{\gamma}_p(t) \cdot f = Y_{\gamma_p(t)} \cdot f = -1.$$

So h is a smooth function with $c := h(0) = f(p) \in [a, b]$ and $h'(t) = -1$ as long as $h(t) \in [a, b]$. We claim that $h(c - a) = a$. To see this let

$$s := \sup\{t \in [0, c - a] \mid h([0, t]) \subseteq [a, b]\}.$$

Assume $s < c - a$. For continuity reasons we get that $h([0, s]) \subseteq [a, b]$ and thus $h'(t) = -1$ for $t \in [0, s]$. Then $h(t) = c - t$ for $t \in [0, s]$. But then $h(s) \in (a, b]$ and h is strictly decreasing around $t = s$ such that there exists $\varepsilon > 0$ with $h([0, s + \varepsilon]) \subseteq [a, b]$, contradicting the definition of s . So $s = c - a$ and we get that $h(t) = c - t$ for $t \in [0, c - a]$. Thus $h(c - a) = a$.

As $h'(t) \leq 0$ for all $t \in \mathbb{R}$ we conclude

$$f(\theta_{b-a}(p)) = h(b - a) \leq h(c - a) = a,$$

thus $\theta_{b-a}(p) \in M_a$. For $p \in M_b \setminus f^{-1}[a, b] = \text{int}M_a$ we also have

$$\theta_{b-a}(p) \in M_a$$

as f decreases along gradient flow lines of Y . This shows (1). We argue similarly for (2). Let $p \in M_a$ and let $\gamma_p: \mathbb{R} \rightarrow M$ be the integral curve of Y starting at p . Consider

$$h: \mathbb{R} \rightarrow \mathbb{R}, \quad t \mapsto f(\gamma_p(-t)).$$

Then $h'(t) = \phi(\gamma_p(-t)) \leq 1$. Assume that $h(b - a) = f(\theta_{a-b}(p)) > b$. Then by the mean value theorem there exists some $\xi \in [0, b - a]$ with

$$h'(\xi) = \frac{h(b - a) - h(0)}{b - a} \geq \frac{h(b - a) - a}{b - a} > 1,$$

which is a contradiction. So $f(\theta_{a-b}(p)) \leq b$, showing (2). Thus θ_{b-a} restricts to a diffeomorphism $M_b \xrightarrow{\cong} M_a$ with inverse θ_{a-b} . $\square$

Counterexample 1.7. Example 1.2 also shows, that we need to impose conditions on the boundary values in Lemma 1.6. To see that we also need that condition for Lemma 1.9 we may similarly define f as the height function on the disjoint union of two cylinders which are slightly shifted in height. Similar examples show, that Theorem 1.10 fails without the assumption on boundary values.

Remark 1.8. We can weaken the assumptions of Lemma 1.6. First of all f does not need to be a Morse function. We can then still construct a vector field X , such that $X \cdot f < 0$ outside of critical points. Secondly, we may replace the assumption that M is compact by instead requiring $f^{-1}[a, b]$ to be compact, which is for example the case if f is proper. We then only have to modify the proof in the definition of U . Instead we define U to be a compact neighbourhood of $K := f^{-1}[a, b]$, i.e. $K \subseteq \text{int}U$, which does not contain any critical points nor boundary points. For example we may define U as a union of finitely many (here we need compactness of $f^{-1}[a, b]$) compact sets K_i not containing neither critical points nor boundary points such that $\text{int}K_i$ still cover K .

Similiarly we may proof the following, for which we can also weaken the assumptions as described in the remark above.

Lemma 1.9. *Let M be a compact manifold with boundary and $f: M \rightarrow \mathbb{R}$ a Morse function. Furthermore let $a, b \in \mathbb{R}$ be regular values of f , such that $[a, b]$ does not contain any critical values nor any boundary values. Then M_a is a (not necessarily smooth) deformation retract of M_b .*

Proof. We consider the flow θ as in Lemma 1.6 and define a retraction

$$r: M_b \rightarrow M_a, p \mapsto \begin{cases} \theta(f(p) - a, p), & f(p) \geq a \\ p, & f(p) \leq a \end{cases}$$

As we have shown above $\theta(f(p) - a, p) \in f^{-1}(a)$. It follows that r is continous and well-defined. To see that r is a deformation retract consider

$$H: M_b \times [0, 1] \rightarrow M_b, (p, t) \mapsto \begin{cases} \theta(t(f(p) - a), p), & f(p) \geq a \\ p, & f(p) \leq a \end{cases}$$

Then $H(-, 0) = id_{M_b}$ and $H(-, 1) = r$. □

We would now also like to generalize the result, that the topology of M changes by attaching a λ -cell when crossing a critical point c_0 of index λ to compact manifolds with boundary. This statement can now be shown using the two lemmas above by requiring $f^{-1}[a, b]$ to only contain the critical point c_0 and no boundary points. The proof now works exactly the same as for the boundaryless case, see [?]. Let us state the theorem for manifolds with boundary:

Theorem 1.10. *Let M be a compact manifold with boundary and $f: M \rightarrow \mathbb{R}$ a Morse function. Let $c_0 \in M$ be a critical point of f of index λ and $c := f(c_0)$ as well as $\varepsilon > 0$, such that $f^{-1}[c - \varepsilon, c + \varepsilon]$ contains no boundary points and no critical points besides c . Then for ε sufficiently small $M_{c+\varepsilon}$ has the homotopy type of $M_{c-\varepsilon}$ with a λ -cell attached. More precisely, there exists an attaching map*

$$\phi: S^{\lambda-1} = \partial D^\lambda \rightarrow \partial M_{c-\varepsilon}$$

and a homotopy equivalence

$$r: M_{c+\varepsilon} \rightarrow M_{c-\varepsilon} \cup_\phi D^\lambda$$

with homotopy inverse

$$\iota: M_{c-\varepsilon} \cup_\phi D^\lambda \rightarrow M_{c+\varepsilon},$$

such that $r \circ \iota = id_{M_{c-\varepsilon} \cup_\phi D^\lambda}$.

Proof Sketch. Following [?] we first construct $F: M \rightarrow \mathbb{R}$, such that

$$F^{-1}(-\infty, c + \varepsilon] = f^{-1}(-\infty, c + \varepsilon] = M_{c+\varepsilon}.$$

Furthermore $F^{-1}[c - \varepsilon, c + \varepsilon]$ does not contain any critical points or boundary points, such that $F^{-1}(-\infty, c - \varepsilon]$ is a deformation retract of $F^{-1}(-\infty, c + \varepsilon]$ by Lemma 1.9. We then show that $M_{c-\varepsilon} \cup_\phi D^\lambda$ is a deformation retract of $F^{-1}(-\infty, c - \varepsilon]$ (or more precisely the image of $M_{c-\varepsilon} \cup_\phi D^\lambda$ under an embedding is a deformation retract). This then shows that $M_{c-\varepsilon} \cup_\phi D^\lambda$ is a deformation retract of $M_{c+\varepsilon}$. □

However this does not yield a cell decomposition of M as we do not know how the topology changes when crossing boundary points. Instead we can use the fact that any compact manifold without boundary admits a cell-decomposition directly, to proof the case when M has non-empty boundary.

Theorem 1.11 (Cell-decompositions of Manifolds). *Let M be compact manifold with or without boundary. Then M has the homotopy type of a CW-complex.*

Proof. The case when M has no boundary is shown in [?]. If $\partial M \neq \emptyset$ we can argue as follows: consider the double $D(M)$ of M . Then $D(M)$ is a compact manifold without boundary. Now we construct a Morse function on $D(M)$. Choose a Morse function $f: M \rightarrow \mathbb{R}$ with $f \leq 0$ and $f(p) = 0$ if and only if $p \in \partial M$, such that f has no critical points on the boundary. Such Morse functions always exist, see [?]. We can then glue f and $-f$ together to get a Morse function F on $D(M)$, again see [?]. Then $M \cong F^{-1}(-\infty, 0]$ and 0 is a regular value of F , allowing us to refer back to the boundaryless case, which tells us that every sublevel set M_a for a regular value $a \in \mathbb{R}$ has the homotopy type of a CW-complex. □

Remark 1.12. We may also formulate Theorem 1.10 and 1.11 for handle bodies instead of CW-complexes. The proof is essentially the same but instead of constructing a deformation retract $F^{-1}(-\infty, c-\varepsilon] \rightarrow M_{c-\varepsilon} \cup_{\phi} D^{\lambda}$ one constructs a homeomorphism $F^{-1}(-\infty, c-\varepsilon] \rightarrow M_{c-\varepsilon} \cup_{\phi} D^{\lambda} \times D^{n-\lambda}$.

1.3 Integral curves and flows

For manifolds with boundary we essentially define integral curves in the same way as for boundaryless manifolds, but allow the domain to be any non-trivial interval (i.e. not a single point) containing 0. In particular this interval may only be half-open or closed. In general, the existence of integral curves on manifolds with boundary fails as illustrated by the following example:

Counterexample 1.13. Consider the lower halfspace $M := \mathbb{R}_{y \leq 0}^2$ and the vector field $X := \frac{\partial}{\partial x} + 2x \frac{\partial}{\partial y}$. Then for $p = (0, 0)$ the integral curve is only defined for $t = 0$, since on $\mathbb{R}^2$ it would be given by $\gamma(t) = (t, t^2)$ but $t^2 > 0$ for $t \neq 0$.

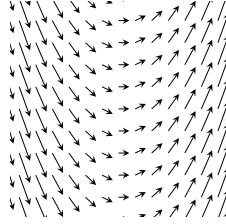


Figure 1: The vector field X . The integral curves are parabolas.

We do however still get a uniqueness result *if* the integral curve exists. Furthermore, if the vector field is sufficiently well-behaved we can guarantee the existence.

Theorem 1.14 (Uniqueness and existence of Integral curves). *Let M be a manifold with boundary, X a smooth vector field on M and $p \in M$. If there exists an integral curve starting at p , then there exists a unique maximal integral curve $\gamma: I \rightarrow M$ starting at p , where one of the four following cases holds for $a \leq 0 \leq b$, $a < b$:*

- (i) $I = (a, b)$
- (ii) $I = [a, b)$ and $\gamma(a) \in \partial M$
- (iii) $I = (a, b]$ and $\gamma(b) \in \partial M$
- (iv) $I = [a, b]$ and $\gamma(a), \gamma(b) \in \partial M$

Furthermore, if X is either transversal to the boundary or everywhere tangent, the unique maximal integral curve starting at p always exists.

Proof. Let's begin by showing uniqueness. The proof is essentially the same as in the boundaryless case (see for example [?]): We show that two integral curves $\gamma: I \rightarrow M$ and $\gamma': I' \rightarrow M$ starting at p agree on their common domain. This then allows us to define the maximal integral curve on some interval I and every interval containing 0 is of the form (i) – (iv)². The only difference is that I and I' may not be open intervals. So we have to show that uniqueness still applies. We claim: Let $U \subseteq \mathbb{R}^n$ be open, $F: U \rightarrow \mathbb{R}^n$ a smooth vector field, I an interval of the form (i) – (iv) and $\gamma, \gamma': I \rightarrow U$ integral curves both starting at $p \in U$. Then $\gamma = \gamma'$. If 0 is an interior point of I we can apply the usual uniqueness theorem for γ, γ' restricted to $\text{int}(I)$. By continuity they must then also agree at the endpoint. So it remains to show uniqueness if 0 is a boundary point of I (say for example $I = [0, b)$). But by the existence theorem for integral curves we can find an integral curve $\eta: (-\varepsilon, \varepsilon) \rightarrow U$ starting at p and define

$$\beta: (-\varepsilon, b) \rightarrow U, t \mapsto \begin{cases} \eta(t), & t \leq 0 \\ \gamma(t), & t \geq 0 \end{cases}$$

Since $\eta(0) = \gamma(0)$ and $\dot{\eta}(0) = F(p) = \dot{\gamma}(0)$ we get that β is at least C^1 . Similarly we define

$$\beta': (-\varepsilon, b) \rightarrow U, t \mapsto \begin{cases} \eta(t), t \leq 0 \\ \gamma'(t), t \geq 0 \end{cases}$$

By the uniqueness theorem we have

$$\beta = \beta'$$

and our claim follows. The case $I = (a, 0]$ goes analogous and for $I = [0, b]$ or $I = [a, 0]$ it is either trivial (if $a = b = 0$) or we consider $I \setminus \{b\}$ resp. $I \setminus \{a\}$ and apply the above argument. Equality at $t = a, t = b$ respectively then again follows by continuity.

To see that $\gamma(t) \in \partial M$ for $t \in \{a, b\}$ a boundary point of I we note that for $p = \gamma(t) \in \text{int}(M)$ we can define the integral curve starting at p on some open interval $(-\varepsilon, \varepsilon)$ by the existence theorem for integral curves (Picard-Lindelöf). But then t is not a boundary point of I since I is maximal.

Next we show the existence if X is transversal to ∂M . For $p \in \text{int}(M)$ this follows directly from the Picard-Lindelöf Theorem as in the boundaryless case. If $p \in \partial M$, we may argue similarly: Consider a chart $h: \mathbb{R}_{x_n \geq 0}^n \supseteq U \rightarrow M$ around $p = h(0)$ and let Y be the vector field X in this chart, i.e. $D_q h(Y_q) = X_{h(q)}$. Then by definition of smoothness we can extend Y to a smooth vector field Y' on some open neighbourhood $U' \subseteq \mathbb{R}^n$ around $0 = h^{-1}(p)$. Applying Picard-Lindelöf we get the existence of an integral curve $\gamma: (-\varepsilon, \varepsilon) \rightarrow U'$ of Y' starting at 0 . Write

$$Y' = \sum_i F_i \frac{\partial}{\partial x_i}$$

and

$$\gamma = (\gamma_1, \dots, \gamma_n).$$

Assume that X is inwards pointing along the boundary. After restricting to a smaller domain we may assume $F_n \geq 0$. Then

$$\gamma'_n(t) = F_n(\gamma(t)) \geq 0,$$

such that γ_n is increasing. In particular $\gamma_n(t) \geq 0$ for $t \in [0, \varepsilon)$, hence $\gamma(t) \in U$ for $t \in [0, \varepsilon)$. Then

$$\eta := h \circ \gamma: [0, \varepsilon) \rightarrow M$$

is the desired integral curve.

For the case that X is everywhere tangent to ∂M see [?] (Theorem 9.34). □

On compact manifolds we can say more about the maximal Interval I :

Lemma 1.15 (Existence of Integral curves on compact manifolds). *If M is compact in Lemma 1.14 then the maximal interval I has one of the following forms:*

- (i) $I = (-\infty, \infty)$
- (ii) $I = [a, \infty)$ and $\gamma(a) \in \partial M$
- (iii) $I = (-\infty, b]$ and $\gamma(b) \in \partial M$
- (iv) $I = [a, b]$ and $\gamma(a), \gamma(b) \in \partial M$

Proof. Let $\gamma: I \rightarrow M$ be the unique maximal integral curve guaranteed by Lemma 1.14. We will show that any open boundary point of I must be $\pm\infty$. For simplicity let us only consider $I = [a, b)$. The other cases are completely analogous. By compactness we can find a sequence $0 < t_n \nearrow b$ such that $\gamma(t_n)$ converges to a point $q \in M$.

Case 1: $q \in \text{int}(M)$. By Picard-Lindelöf we can find an open neighbourhood $U \subseteq M$ around q and $\varepsilon > 0$, such that for every point $x \in U$ the integral curve starting at x is defined for $t \in (-\varepsilon, \varepsilon)$.

²If we put together integral curves defined on open intervals that agree on their common domain, it is clear that this results in a smooth curve, since around each t this curve is given by one of the smooth curves which we put together. If we put together integral curves γ_1, γ_2 defined on something like $I_1 = (a, b]$ and $I_2 = [b, c)$ we have to be more careful around the boundary point b . But the same argument as we applied to β in the proof of Lemma 1.14 shows that putting γ_1 and γ_2 together results in a C^1 -curve. We can then apply the usual uniqueness theorem to see that this curve is already given by a C^∞ -curve.

For some fixed m sufficiently large we have $q_0 := \gamma(t_m) \in U$. Assume that $b - t_m < \frac{\varepsilon}{2}$. But then we can extend γ to $[a, b + \frac{\varepsilon}{2})$ as follows. Consider

$$\gamma': [a - t_m, b - t_m) \rightarrow M, \quad t \mapsto \gamma(t + t_m).$$

Then γ' is a integral curve starting at q_0 . Let $\alpha: (-\varepsilon, \varepsilon) \rightarrow M$ be another integral curve starting at q_0 . Then α and γ' agree on their common domain by the arguments in Lemma 1.14 and thus we can put them together to define an integral curve

$$\beta: J \rightarrow M$$

for $J := [a - t_m, b - t_m) \cup (-\varepsilon, \varepsilon) \supseteq [a - t_m, \varepsilon)$. But then we can define

$$\gamma'': [a, \varepsilon + t_m) \rightarrow M, \quad t \mapsto \beta(t - t_m),$$

which is an integral curve starting at $\gamma(0) = p$. However

$$b + \frac{\varepsilon}{2} < t_m + \varepsilon$$

and so $[a, b + \frac{\varepsilon}{2}) \subseteq [a, \varepsilon + t_m)$, contradicting maximality of γ .

Case 2: $q \in \partial M$. We argue similarly as above. In a local chart around q we can again apply Picard-Lindelöf (smoothness of X guarantees by definition the existence of a smooth extension of X to an open neighbourhood in $\mathbb{R}^n$). We then again take $q_0 := \gamma(t_m)$ sufficiently close to q , choose an integral curve α starting at q , which we can use to define an integral curve α' starting at q_0 defined on some open interval $(-\varepsilon, \varepsilon)$ by translation in time as above. Also we define γ' as above. The same argument as above then shows that we can extend γ' (in the local chart) to an integral curve β on $J := [a - t_m, b - t_m) \cup (-\varepsilon, \varepsilon) \supseteq [a - t_m, b - t_m]$. In particular we have $\beta([a - t_m, b - t_m]) \subseteq M$. Translating in time again yields an integral curve $\gamma'': [a, b] \rightarrow M$ contradicting maximality. $\square$

As an application we can show:

Lemma 1.16. *Let I be some proper interval containing 0 and $\gamma: I \rightarrow M$ any smooth regular curve, i.e. $\dot{\gamma}(t) \neq 0$ for all $t \in I$. Then there exists a unique unit speed reparametrization of γ , that is an orientation preserving diffeomorphism $\phi: J \rightarrow I$ with $\phi(0) = 0$, such that $\gamma \circ \phi$ is a unit speed curve.*

Proof. We have

$$(\gamma \circ \phi)'(t) = \phi'(t) \dot{\gamma}(\phi(t)),$$

so we need to find ϕ , such that (note that we also need $\phi' > 0$ to preserve orientation)

$$\phi'(t) = \frac{1}{|\dot{\gamma}(\phi(t))|}$$

This is a flow equation for the smooth vector field

$$X(p) := \frac{1}{|\dot{\gamma}(p)|}$$

on I . Since X is transversal to the boundary of I (if I even has boundary), Theorem 1.14 yields the existence of a unique maximal integral curve $\phi: J \rightarrow I$ of X with $\phi(0) = 0$. We now only need to show that ϕ is surjective, since $\phi' > 0$ already shows that it is an embedding.

To this end, consider any compact subinterval $I' := [a, b] \subseteq I$ ($a < b$) containing 0 and let $\psi: J' \rightarrow I'$ be the unique maximal integral curve of $X|_{I'}$ starting at 0. By Lemma 1.15 ψ has to be defined on a compact interval $J' = [c, d]$, since $X|_{I'}$ is bounded below by a positive constant, so if J' would be infinite in the positive/negative direction, ψ would also tend to positive/negative infinity, but by definition it is contained in the compact interval I' . Moreover Lemma 1.15 then yields

$$\psi(c) = a, \quad \psi(d) = b.$$

Since ϕ is the unique maximal integral curve of X we get that $J' \subseteq J$ and $\phi|_{J'} = \psi$. In particular $I' \subseteq \text{im}(\phi)$. This completes the proof. $\square$

Remark 1.17 (A note on flows for manifolds with boundary). In general the concept of a flow is not well behaved if M has boundary, as integral curves may not even exist (see Counterexample 1.13). In [?] the author shows, that if X is tangent to ∂M we still get a well-behaved flow, i.e. it is defined on some open subset $\mathcal{O} \subseteq M \times \mathbb{R}$. I think that if we instead assume, that X is transversal to ∂M , we still get a "good" flow domain in the sense that we can define the flow on some subset $\mathcal{D} \subseteq M \times \mathbb{R}$, which is a codimension 0 submanifold with boundary and $I_p := \{t \in \mathbb{R} \mid (p, t) \in \mathcal{D}\}$ is an interval around 0 for each $p \in M$.

The statement in the above remark should follow from the following lemma, which allows us to define submanifold charts for $\mathcal{D}$.

Lemma 1.18. *Let M be a manifold with boundary and X a vector field of M , which is inward pointing along ∂M . Let*

$$M_{\partial} := \{p \in M \mid \text{im} \gamma_p \cap \partial M \neq \emptyset\}$$

be all points in M whose trajectory contains boundary points (this is at most one for each $p \in M$). For $p \in M_{\partial}$ the integral curve γ_p starting at p is defined on some interval $[-\tau(p), a)$ for some $\tau(p) \geq 0$ and $a > -\tau(p)$. Then the following statements hold true

- (i) M_{∂} is an open subset of M
- (ii) $\tau: M_{\partial} \rightarrow \mathbb{R}_{\geq 0}, p \mapsto \tau(p)$ is smooth

Proof sketch. The idea is to consider the maximal flow

$$\theta: \mathcal{D} \rightarrow M, (p, t) \mapsto \gamma_p(t)$$

on some open subset $\mathcal{D} \subseteq \partial M \times [0, \infty)$ and show that this is an embedding. Then

$$M_{\partial} = \text{im} \theta$$

is open and

$$\tau = \pi \circ \theta^{-1}: M_{\partial} \rightarrow [0, \infty)$$

where $\pi: \partial M \times [0, \infty) \rightarrow [0, \infty)$ is the canonical projection map. □

Lemma 1.19. *Injective Flowouts are already embeddings*